Formula & Decision Chart

- **CORE FORMULAS**
- $P = F / (1+i)^n$
- $F = P(1+i)^n$
- $P = A(P/A,i,n)$ $A = P(A/P,i,n)$
- $AW = PW \cdot (A/P,i,n)$
- $PW = \Sigma\ CF_t / (1+i)^t$
- IRR: $0 = \Sigma\ CF_t / (1+IRR)^t$
- ΔIRR: $0 = PW(B - A)$
- B/C = PW(Benefits) / PW(Costs)

- **DECISION RULES**
- • Independent project: PW ≥ 0 or IRR ≥ MARR → Accept
- • Mutually exclusive: Use PW or AW or Incremental IRR
- • Different lives: Use AW
- • Non-conventional CF: Use PW or EROR
- • Public projects: Use B/C (or Incremental B/C)

- **QUICK FLOW**
- Public? → B/C
- Mutually exclusive? → PW / AW / ΔIRR
- Different lives? → AW
- Otherwise → PW or IRR

1) Time Value of Money — Core Factors

$$P = F(P/F, i, n) = \frac{F}{(1+i)^n}$$

$$F = P(F/P, i, n) = P(1+i)^n$$

$$A = P(A/P, i, n) = P\frac{i(1+i)^n}{(1+i)^n - 1}$$

$$P = A(P/A, i, n) = A\frac{(1+i)^n - 1}{i(1+i)^n}$$

$$A = F(A/F, i, n) = F\frac{i}{(1+i)^n - 1}$$

2) Present Worth (PW)

$$PW(i) = \sum_{t=0}^{n} \frac{CF_t}{(1+i)^t}$$

Decision:

- If **PW ≥ 0** → Accept
- If **PW < 0** → Reject
- For alternatives: **Choose largest PW**

3) Annual Worth (AW)

$$AW = PW \cdot (A/P, i, n)$$

Or directly:

$$AW = \text{Annual benefits} - \text{Annual costs}$$

Decision:

- If **AW ≥ 0** → Accept
- Choose alternative with **largest AW**

Best for **different lives**

4) Internal Rate of Return (IRR / ROR)

$$0 = \sum_{t=0}^{n} \frac{CF_t}{(1 + IRR)^t}$$

Decision (Single Project):

- If **IRR ≥ MARR** → Accept
- If **IRR < MARR** → Reject

IRR is only a screening tool for mutually exclusive projects!

5) Incremental IRR (ΔIRR)

1. Compute:

$$\Delta CF = B - A$$

2. Solve:

$$0 = PW(\Delta CF)$$

Decision:

- If **ΔIRR ≥ MARR** → Choose **higher-cost** alternative
- If **ΔIRR < MARR** → Choose **lower-cost** alternative

6) External Rate of Return (EROR) / MIRR

Used when:

- Non-conventional cash flows
- Multiple IRR problem

Logic:

- Positive cash flows → reinvest at **investment rate**
- Negative cash flows → borrow at **borrowing rate**

Decision:

- If **EROR ≥ MARR** → Accept

7) Benefit–Cost Ratio (Public Projects)

$$B/C = \frac{PW(\text{Benefits})}{PW(\text{Costs})}$$

Disbenefits are **subtracted from benefits**

Decision:

- If **B/C ≥ 1** → Accept
- For alternatives: use **Incremental B/C**

8) Which Method Should I Use?

Situation	Use
Independent projects	IRR or PW
Mutually exclusive	PW / AW / ΔIRR
Different lives	AW
Non-conventional CF	PW / EROR
Capital intensive	PW
Public projects	B/C

EXAMPLE 1

A project has the following cash flows. MARR = 8%.

Year	Cash Flow ($)
0	−8,000
1	+2,500
2	+3,000
3	+3,000

(a) Write the PW equation.

(b) Substitute the numerical values.

(c) Compute PW.

Cash flows: $CF_0 = -8000,\ CF_1 = 2500,\ CF_2 = 3000,\ CF_3 = 3000$

(a) PW equation

$$PW = -8000 + \frac{2500}{(1.08)^1} + \frac{3000}{(1.08)^2} + \frac{3000}{(1.08)^3}$$

(b) Compute

$$PW = -8000 + 2314.81 + 2572.02 + 2381.50$$

$$\boxed{PW \approx -731.67}$$

EXAMPLE 2

A server system has:

- $C_0 = -12{,}000$
- Annual net savings: $+3{,}500$ at the end of years 1–5
- Overhaul cost: $-2{,}000$ at the end of year 3
- Salvage value: $+1{,}000$ at the end of year 5

(a) Write PW equation.

(b) Compute PW.

$$PW = -12000 + 3500(P/A, 10\%, 5) - 2000(P/F, 10\%, 3) + 1000(P/F, 10\%, 5)$$

Compute factors:

$$(P/A, 10\%, 5) = \frac{1 - (1.1)^{-5}}{0.1} = 3.79079$$

$$(P/F, 10\%, 3) = (1.1)^{-3} = 0.75131, \quad (P/F, 10\%, 5) = (1.1)^{-5} = 0.62092$$

Now:

$$PW = -12000 + 3500(3.79079) - 2000(0.75131) + 1000(0.62092)$$

$$PW = -12000 + 13267.77 - 1502.63 + 620.92 = 386.06$$

EXAMPLE 3

Two machines are used for the same purpose. MARR = 10%.

Machine A:

- Cost = 10,000
- Life = 4 years
- Salvage = 0
- Annual operating cost = 2,500

Machine B:

- Cost = 14,000
- Life = 6 years
- Salvage = 0
- Annual operating cost = 1,800

(a) Write the AW expression for Machine A.

(b) Write the AW expression for Machine B.

(c) Compute AW(A) and AW(B).

(d) Which machine should be selected?

Machine A

Cost = 10,000; Life = 4; Salvage = 0; Annual O&M = 2,500

$$AW_A = -10000(A/P, 10\%, 4) - 2500$$

$$(A/P, 10\%, 4) = \frac{0.10(1.10)^4}{(1.10)^4 - 1} = 0.31547$$

$$AW_A = -10000(0.31547) - 2500 = -5654.71$$

$$\boxed{AW_A \approx -5654.71}$$

Machine B

Cost = 14,000; Life = 6; Salvage = 0; Annual O&M = 1,800

$$AW_B = -14000(A/P, 10\%, 6) - 1800$$

$$(A/P, 10\%, 6) = 0.22961$$

$$AW_B = -14000(0.22961) - 1800 = -5014.50$$

$$\boxed{AW_B \approx -5014.50}$$

Selection (cost-only): choose the **less negative AW**

$$\boxed{\text{Select Machine B}}$$

EXAMPLE 4

Machine A: cost 9,000; life 3 years; salvage 1,000; annual O&M 2,200

Machine B: cost 13,000; life 5 years; salvage 2,000; annual O&M 1,700

(a) Write AW(A), AW(B)

(b) Compute and select.

General:

$$AW = -P(A/P, i, n) + S(A/F, i, n) - \text{O\&M}$$

A:

$$AW_A = -9000(A/P, 8\%, 3) + 1000(A/F, 8\%, 3) - 2200$$

Factors:

$$(A/P, 8\%, 3) = \frac{0.08(1.08)^3}{(1.08)^3 - 1} = 0.38803$$

$$(A/F, 8\%, 3) = \frac{0.08}{(1.08)^3 - 1} = 0.30803$$

So:

$$AW_A = -9000(0.38803) + 1000(0.30803) - 2200$$

$$= -3492.27 + 308.03 - 2200 = -5384.24$$

$$\boxed{AW_A \approx -5384.24}$$

B:

$$AW_B = -13000(A/P, 8\%, 5) + 2000(A/F, 8\%, 5) - 1700$$

Factors:

$$(A/P, 8\%, 5) = 0.25046, \quad (A/F, 8\%, 5) = 0.17046$$

So:

$$AW_B = -13000(0.25046) + 2000(0.17046) - 1700$$

$$= -3255.98 + 340.92 - 1700 = -4615.06$$

$$\boxed{AW_B \approx -4615.06}$$

Select less negative annual cost:

$$\boxed{\text{Select Machine B}}$$

EXAMPLE 5

Two design options:

- **A:** $CF_0 = -15{,}000;\ CF_{1-4} = +6{,}000$
- **B:** $CF_0 = -20{,}000;\ CF_{1-4} = +7{,}200$

(a) Compute incremental cash flow $B - A$.

(b) Compute $PW(\Delta)$.

(c) Decide.

Incremental:

$$\Delta CF_0 = -20000 - (-15000) = -5000$$

$$\Delta CF_{1-4} = 7200 - 6000 = +1200$$

$$PW(\Delta) = -5000 + 1200(P/A, 12\%, 4)$$

$$(P/A, 12\%, 4) = \frac{1 - (1.12)^{-4}}{0.12} = 3.03735$$

$$PW(\Delta) = -5000 + 1200(3.03735) = -5000 + 3644.82 = -1355.18$$

$$\boxed{PW(\Delta) \approx -1355.18}$$

Since incremental PW is negative, do **not** pay extra for B:

$$\boxed{\text{Select A}}$$

EXAMPLE 6

A project has the following cash flows:

Year	Cash Flow ($)
0	−4,000
1	+2,000
2	+2,000
3	+2,000

(a) Write the IRR equation.

(b) Solve for IRR (show at least two trials).

(c) If MARR = 10%, state your decision.

Cash flows: $CF_0 = -4000,\ CF_1 = CF_2 = CF_3 = 2000$

(a) IRR equation

$$0 = -4000 + \frac{2000}{(1+i)} + \frac{2000}{(1+i)^2} + \frac{2000}{(1+i)^3}$$

(b) Two trials

- Trial 1: $i = 20\%$

$$PW = -4000 + 2000(P/A, 20\%, 3)$$

$$(P/A, 20\%, 3) = \frac{1 - (1.2)^{-3}}{0.2} = 2.10648 \Rightarrow PW$$

$$= -4000 + 2000(2.10648) = +212.96$$

- Trial 2: $i = 25\%$

$$(P/A, 25\%, 3) = 1.95200 \Rightarrow PW = -4000 + 2000(1.95200) = -96.00$$

So IRR is between 20% and 25%. Linear interpolation:

$$IRR \approx 20\% + \frac{212.96}{212.96 + 96.00} \cdot 5\% = 23.45\%$$

$$\boxed{IRR \approx 23.45\%}$$

(c) Decision

$$\boxed{23.45\% > 10\% \Rightarrow \text{Accept}}$$

EXAMPLE 7

Two alternatives are given. MARR = 10%.

Year	A ($)	B ($)
0	−8,000	−11,000
1–4	+3,000	+4,200

(a) Construct the incremental cash flow (B − A).

(b) Write the ΔIRR equation.

(c) Compute ΔIRR and state your decision.

A: $CF_0 = -8000$, $CF_{1-4} = +3000$

B: $CF_0 = -11000$, $CF_{1-4} = +4200$

(a) Incremental cash flow (B – A)

$$\Delta CF_0 = -11000 - (-8000) = -3000$$

$$\Delta CF_{1-4} = 4200 - 3000 = +1200$$

(b) ΔIRR equation

$$0 = -3000 + 1200(P/A, i, 4)$$

$$1200(P/A, i, 4) = 3000 \Rightarrow (P/A, i, 4) = 2.5$$

(c) Solve for i

Find i such that $(P/A, i, 4) = 2.5$. This gives:

$$\boxed{\Delta IRR \approx 21.86\%}$$

Decision:

$$\boxed{21.86\% > 10\% \Rightarrow \text{Select B}}$$

EXAMPLE 8

Cash flows:

Year	CF
0	−10,000
1	+14,000
2	−5,000

(a) Explain why multiple IRRs may exist (sign changes).

(b) Evaluate PW at MARR = 10% and decide.

Sign changes: $-, +, -$ → **two sign changes** → possibly **0/1/2 IRRs**.

PW at 10%:

$$PW = -10000 + \frac{14000}{1.1} - \frac{5000}{1.1^2}$$

$$PW = -10000 + 12727.27 - 4132.23 = -1404.96$$

$$\boxed{PW \approx -1404.96 \Rightarrow \text{Reject at } 10\%}$$

EXAMPLE 9

For a project, the present worth (PW) has been computed at different interest rates:

i	PW
6%	+900
10%	+200
14%	−700

Assume the reinvestment rate is 9%.

(a) Write the EROR condition (PW = 0).

(b) Determine the interval in which EROR lies.

(c) Estimate EROR by linear interpolation.

Given:

i	PW
6%	+900
10%	+200
14%	−700

(a) Condition

$$PW(i) = 0$$

(b) Bracket

PW changes sign between 10% and 14%:

$$\boxed{10\% < EROR < 14\%}$$

(c) Linear interpolation

$$EROR \approx 10\% + \frac{200}{200 + 700}(14\% - 10\%)$$

$$EROR \approx 10\% + \frac{200}{900} \cdot 4\% = 10.8889\%$$

$$\boxed{EROR \approx 10.89\%}$$

EXAMPLE 10

Given PW values:

i	PW
5%	+600
9%	+120
11%	−80

(a) Find the interval for EROR.

(b) Estimate EROR by linear interpolation.

Sign change between 9% and 11%:

$$\boxed{9\% < EROR < 11\%}$$

Interpolation:

$$EROR \approx 9\% + \frac{120}{120 + 80}(11\% - 9\%)$$

$$EROR \approx 9\% + \frac{120}{200} \cdot 2\% = 9\% + 1.2\% = 10.2\%$$

$$\boxed{EROR \approx 10.20\%}$$